

HAT-P-54b

R-1549

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_191108 · created 2026-08-01T19:49:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458795.8097 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.106 +/- 0.045
Transit depth (Rp/Rs)^2	1.12 +/- 0.95 [%]
Orbital Inclination (inc)	86.94 +/- 2.18
Airmass coefficient 1 (a1)	0.95 +/- 0.025
Airmass coefficient 2 (a2)	0.0329 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	2.62 %
Best Comparison Star	#1 - [232, 80]
Optimal Method	PSF photometry
Transit Duration (day)	0.061 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.106 ± 0.045 vs 0.1572 (deviation 1.14σ)
Duration fit vs published	0.061 d vs 0.075 d (deviation 1.27σ)
Mid-time O–C	2.0 min
Depth SNR	2.4
Residual scatter	2.62 %

Light curve

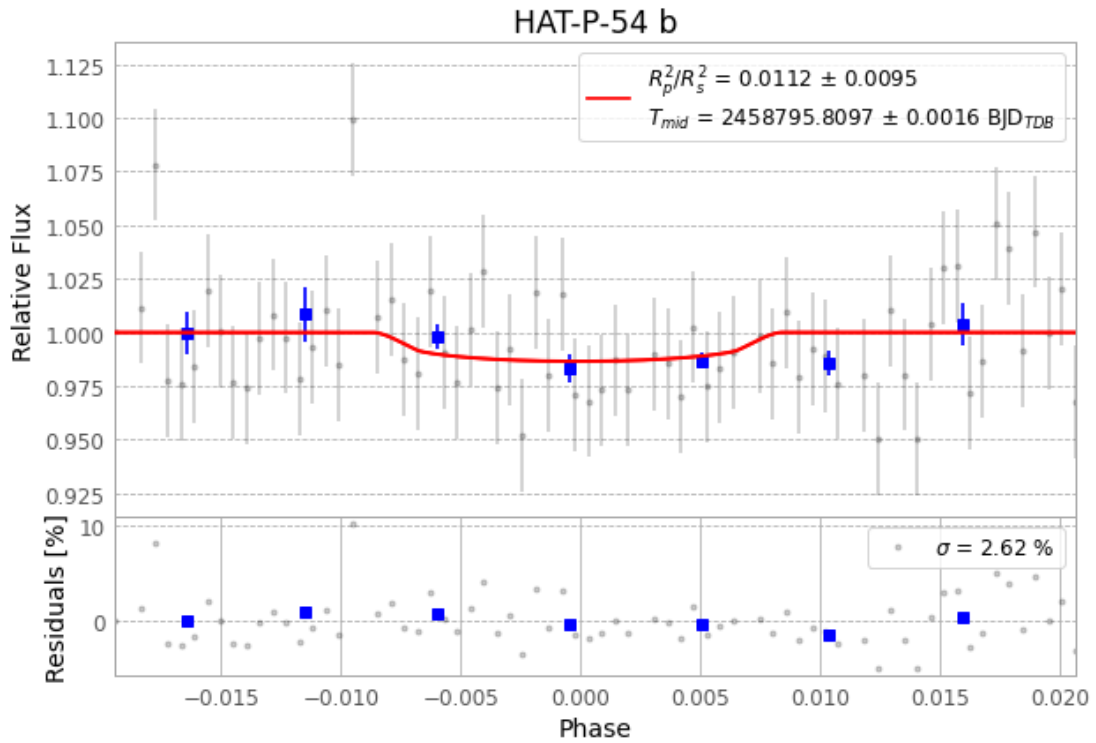


Figure 1 — FinalLightCurve_HAT-P-54 b_2019-11-07.png (SHA-256 752d88adc7203657...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [244, 155], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 68abe6f65f409ee0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), HATP-54191108053312.FITS → HATP-54191108091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-191108.log (b249723cc11c...), FinalLightCurve_HAT-P-54 b_2019-11-07.png (752d88adc720...), FinalLightCurve_HAT-P-54 b_2019-11-07.csv (34ae811753e8...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 19:49Z.